

A Convex-Order Solution of the Generalized-Cone Conjecture

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Abstract

González Merino reduced a family of sharp Hermite–Hadamard inequalities to a one-parameter maximization over truncated cones and conjectured that the maximum is always attained by a cone. We prove the conjectured inequality in every dimension. The axial volume law of a truncated cone, normalized by its median, has an explicit quantile. Its quantile crosses the cone quantile exactly once, and its mean is no larger. These two facts give increasing-convex-order domination by the cone law, simultaneously for every admissible convex function. The mean estimate is the only nonlocal step; it follows from concavity on one side of the parameter space and a positive-coefficient power-series ratio on the other. For strictly convex functions the maximizer is unique in dimensions at least two, yielding the proposed equality case under the intended nonzero normalization.

1 The conjecture and the result

Let $C \subset \mathbb{R}^n$ be a convex body and let $f : C \rightarrow [0, \infty)$ be concave. The source associates with (C, f) a point $x_{C,f}$ whose definition is based on a volume-halving section. Its Conjecture 1.7, reproduced in Figure 1, asks whether, for every convex $\phi : [0, \infty) \rightarrow [0, \infty)$ with $\phi(0) = 0$,

$$\frac{1}{|C|} \int_C \phi(f(x)) dx \leq n \int_0^1 \phi \left(f(x_{C,f}) \frac{2^{1/n} t}{2^{1/n} - 1} \right) (1-t)^{n-1} dt. \quad (1)$$

Theorem 1 (Full resolution of the inequality). *Inequality (1) holds in every dimension $n \geq 1$. If $n \geq 2$, ϕ is strictly convex, and $f(x_{C,f}) > 0$, equality holds if and only if C is a generalized cone and f is affine and vanishes on its nondegenerate base.*

The positivity condition in the equality statement is necessary. As the source conjecture is literally phrased, $f \equiv 0$ gives equality for every convex body C , including bodies that are not generalized cones. The paper’s optimization problem is introduced with the normalization $f(x_{C,f}) = k > 0$; Theorem 1 proves exactly that intended nontrivial equality statement.

2 Proof intuition

The source has already done the geometric work: its main theorem replaces an arbitrary pair (C, f) by an affine function on a truncated cone. What remains is to compare a one-parameter family of axial probability laws. Normalize each affine function at the volume median. The resulting law has an explicit quantile R_a .

After deriving the solutions of Theorems [1.2](#), [1.4](#), [1.5](#) and [1.6](#), it is natural to conjecture that the maximum in the right-hand side in Theorem [1.1](#) is attained when C is a cone, and f is an affine function becoming 0 at its base, thus yielding the following *general result*:

Conjecture 1.7. *Let $C \in \mathcal{K}^n$, $f : C \rightarrow [0, \infty)$ be concave, and $\phi : [0, \infty) \rightarrow [0, \infty)$ be convex with $\phi(0) = 0$. Then*

$$\frac{1}{|C|} \int_C \phi(f(x)) dx \leq n \int_0^1 \phi \left(f(x_{C,f}) \frac{2^{1/n} t}{2^{1/n} - 1} \right) (1-t)^{n-1} dt.$$

Moreover, if ϕ is strictly convex, equality holds if and only if C is a generalized cone, and f is an affine function which becomes zero at the base of C .

Figure 1: Conjecture 1.7 in arXiv:2005.13839v2, PDF p. 5.

Two elementary-looking comparisons finish the problem. First, R_a and the cone quantile R_0 cross exactly once, at their common median 1. This follows by differentiating with respect to the truncation parameter; the derivative sign is controlled by the decreasing elasticity of $(1+z)^{1/n} - 1$. Second, R_0 has the largest mean. On one half of the parameter range this is the midpoint inequality for the concave function $x^{1/n}$. On the other half it becomes monotonicity of a ratio of two positive power series.

The single crossing controls stop-loss functions above the median. Below the median, rewrite a stop-loss as the mean plus a put; the mean comparison and the lower quantile crossing control the two terms. Thus the cone law dominates in increasing convex order, which is precisely the simultaneous inequality for every admissible ϕ .

3 Reduction to a normalized axial law

We use the source's Theorem 1.1 [1, Theorem 1.1]. It states that the left side of (1) is no larger than the maximum obtained on generalized truncated cones R , with an affine function g which vanishes on one base and satisfies

$$g(x_{R,g}) = f(x_{C,f}).$$

Moreover, for strictly convex ϕ , equality in this reduction forces C itself to be one of the maximizing generalized truncated cones and f to be affine and zero on the corresponding base. It therefore suffices to solve the remaining one-parameter maximization.

Normalize the axis of a truncated cone to $[0, 1]$, put the zero-value base at $t = 0$, and scale its two base radii to 1 and $q \geq 0$. The case in which the first radius is zero is included as $q = \infty$. Under normalized volume measure, the axial coordinate T_q has density proportional to

$$(1 + (q-1)t)^{n-1}, \quad 0 \leq t \leq 1.$$

Put

$$r = \frac{1}{n}, \quad a = q^n.$$

For $q \neq 1$, the distribution function and quantile of T_q are

$$F_q(t) = \frac{(1 + (q-1)t)^n - 1}{q^n - 1}, \quad (2)$$

$$Q_a(u) = \frac{(1 + (a-1)u)^r - 1}{a^r - 1}. \quad (3)$$

The formulas at removable values are understood by continuity.

The point $x_{R,g}$ is on the volume-halving section, so its axial coordinate is the median $Q_a(1/2)$. If $c = f(x_{C,f})$, then $g(t) = ct/Q_a(1/2)$. Thus

$$\frac{1}{|R|} \int_R \phi(g(x)) dx = \mathbb{E}\phi(cX_a), \quad X_a := \frac{T_q}{Q_a(1/2)}.$$

By (3), the quantile of X_a is

$$R_a(u) = \frac{(1 + (a-1)u)^r - 1}{((1+a)/2)^r - 1}. \quad (4)$$

In particular, the cone with nondegenerate zero-value base corresponds to $a = 0$, and

$$R_0(u) = \frac{1 - (1-u)^r}{1 - 2^{-r}}. \quad (5)$$

The problem is now purely one-dimensional: prove $X_a \leq_{\text{icx}} X_0$ for every $a \in [0, \infty]$.

4 The quantiles cross once

Lemma 2 (Single crossing). *For every $a \geq 0$,*

$$R_0(u) \leq R_a(u) \quad (0 \leq u \leq \tfrac{1}{2}), \quad R_0(u) \geq R_a(u) \quad (\tfrac{1}{2} \leq u \leq 1).$$

Proof. For $0 < r \leq 1$, set

$$\psi(z) = (1+z)^r - 1, \quad E(z) = \frac{z\psi'(z)}{\psi(z)} = \frac{rz(1+z)^{r-1}}{(1+z)^r - 1}.$$

The removable value at $z = 0$ is understood continuously. With $y = 1 + z$, direct differentiation gives

$$E'(z) = \frac{ry^{r-2}}{(y^r - 1)^2} (y^r - ry - (1-r)) \leq 0. \quad (6)$$

Indeed, concavity of y^r gives $y^r \leq ry + 1 - r$ for every $y \geq 0$.

Set $v = (a-1)/2$ and $\tau = 2u$. Equation (4) becomes

$$R_a(u) = \frac{\psi(\tau v)}{\psi(v)}.$$

Away from removable endpoints,

$$\frac{\partial}{\partial v} \log R_a(u) = \frac{E(\tau v) - E(v)}{v}. \quad (7)$$

If $\tau < 1$, the right side is nonnegative both for $v < 0$ and $v > 0$, because E is decreasing. If $\tau > 1$, it is nonpositive. Hence $R_a(u)$ increases with a below $u = 1/2$ and decreases with a above $u = 1/2$. Comparison with the endpoint $a = 0$ proves the result; continuity handles the removable cases. $\square$

5 The cone also has the largest mean

Lemma 3 (Mean domination). *For every $a \in [0, \infty]$,*

$$M(a) := \mathbb{E}X_a = \int_0^1 R_a(u) du \leq M(0) = \frac{r}{(r+1)(1-2^{-r})}. \quad (8)$$

If $0 < r < 1$, equality occurs only at $a = 0$.

Proof. Suppose first that $a \geq 1$. Integrating (4) shows that

$$M(a) = \frac{\frac{1}{a-1} \int_1^a x^r dx - 1}{((1+a)/2)^r - 1}.$$

Concavity of x^r gives

$$\frac{1}{a-1} \int_1^a x^r dx \leq \left(\frac{1+a}{2} \right)^r,$$

so $M(a) \leq 1$. On the other hand, $M(0) \geq 1$ because $2^r \leq 1+r$ for $0 < r \leq 1$. Both inequalities are strict when $0 < r < 1$ and $a > 1$, while $M(1) = 1 < M(0)$.

It remains to consider $0 \leq a \leq 1$. Put $h = 1 - a$ and

$$p(x) = 1 - (1-x)^r.$$

For $0 < r < 1$ its binomial expansion on $[0, 1]$ is

$$p(x) = \sum_{k \geq 1} c_k x^k, \quad c_k > 0.$$

Equation (4) gives

$$M(1-h) = \frac{\int_0^1 p(hu) du}{p(h/2)} = \frac{\sum_{k \geq 1} c_k h^k / (k+1)}{\sum_{k \geq 1} c_k h^k / 2^k}. \quad (9)$$

This ratio is strictly increasing in h . For completeness, put

$$A(h) = \sum_{k \geq 1} \alpha_k h^k, \quad B(h) = \sum_{k \geq 1} \beta_k h^k, \quad \frac{\alpha_k}{\beta_k} = \rho_k := \frac{2^k}{k+1}.$$

The sequence ρ_k is strictly increasing, and pairing the (i, j) and (j, i) terms gives

$$A'B - AB' = \sum_{j > i} (j-i) \beta_i \beta_j (\rho_j - \rho_i) h^{i+j-1} > 0 \quad (h > 0).$$

Consequently $M(1-h)$ is largest at $h = 1$, equivalently $a = 0$. Continuity gives the endpoint. If $r = 1$, (4) reduces to $R_a(u) = 2u$ for every a , and the asserted non-strict inequality is immediate. $\square$

6 Increasing-convex order and completion

Lemma 4 (Stop-loss domination). *For every $a \geq 0$ and every $s \geq 0$,*

$$\mathbb{E}(X_a - s)_+ \leq \mathbb{E}(X_0 - s)_+.$$

Consequently $X_a \leq_{\text{icx}} X_0$.

Proof. Every X_a has median 1. If $s \geq 1$, quantiles below $1/2$ contribute nothing to either stop-loss expectation, while Lemma 2 orders the upper quantiles in the required direction.

If $0 \leq s \leq 1$, quantiles above $1/2$ contribute nothing to the put expectations, and the lower-half inequality in Lemma 2 gives

$$\mathbb{E}(s - X_a)_+ \leq \mathbb{E}(s - X_0)_+.$$

Using $(x - s)_+ = x - s + (s - x)_+$ and Lemma 3, we obtain

$$\begin{aligned} \mathbb{E}(X_0 - s)_+ - \mathbb{E}(X_a - s)_+ &= \mathbb{E}X_0 - \mathbb{E}X_a \\ &\quad + \mathbb{E}(s - X_0)_+ - \mathbb{E}(s - X_a)_+ \geq 0. \end{aligned}$$

This proves the stop-loss inequalities.

Finally, any finite convex nonnegative function on $[0, \infty)$ satisfying $\phi(0) = 0$ is increasing and has the call-function representation

$$\phi(x) = \alpha x + \int_{[0, \infty)} (x - s)_+ \mu(ds), \quad \alpha \geq 0, \quad \mu \geq 0,$$

on every bounded interval. Integrating the stop-loss inequalities proves $\mathbb{E}\phi(cX_a) \leq \mathbb{E}\phi(cX_0)$ for every $c \geq 0$. $\square$

Proof of Theorem 1. For $a = 0$, the axial coordinate has density

$$n(1 - t)^{n-1}, \quad 0 \leq t \leq 1,$$

and median

$$Q_0(1/2) = 1 - 2^{-1/n} = \frac{2^{1/n} - 1}{2^{1/n}}.$$

Therefore

$$\mathbb{E}\phi(cX_0) = n \int_0^1 \phi\left(c \frac{2^{1/n}t}{2^{1/n} - 1}\right) (1 - t)^{n-1} dt.$$

Lemma 4 proves that this dominates every truncated-cone term in the source's Theorem 1.1. That theorem now gives (1).

Assume $n \geq 2$ and $c > 0$. Lemma 3 is strict for every $a > 0$. In fact the proof of Lemma 4 then gives a strict stop-loss gap for every $0 \leq s \leq 1$. After scaling by c , this gap is strict for $0 < s < c$. The curvature measure of a strictly convex ϕ charges every nonempty interval, so

$$\mathbb{E}\phi(cX_a) < \mathbb{E}\phi(cX_0) \quad (a > 0).$$

Thus the unique maximizing parameter is $a = 0$: the truncated cone is a cone and the affine function vanishes on its nondegenerate base. The equality statement in the source's Theorem 1.1 yields the claimed characterization of (C, f) .

For $n = 1$, $R_a(u) = 2u$ for every a . The scalar law is independent of the truncation parameter, and one-dimensional convex bodies are intervals, which are generalized cones under the natural zero-dimensional-base convention. $\square$

7 Consistency and novelty notes

The source’s three-dimensional constant. Putting $\phi(t) = t$ in Conjecture 1.7 gives the sharp coefficient

$$\frac{2^{1/n}}{(n+1)(2^{1/n} - 1)}.$$

For $n = 3$ this is one twelfth of the coefficient printed in the source’s Theorem 1.6. The discrepancy is visible in the source proof: the integral of $t(r + mt)^2$ is expanded without its denominator 12. The conjecture, the planar formula, and the direct cone computation all agree with the coefficient above. This arithmetic issue is downstream of Theorem 1.1 and does not enter the reduction used here.

Literature search. The arXiv version and the final 2021 journal version both state the target as Conjecture 1.7. Bounded exact-title, formula-fragment, and core-keyword searches performed on August 9, 2026 found no later proof or counterexample. Crossref and OpenAlex both reported zero citing works for the published article (DOI 10.1007/s10231-021-01080-y). This supports, but cannot certify, novelty; specialist review remains appropriate.

References

- [1] B. González Merino, *Estimating the average of functions with convexity properties by means of a new center*, Ann. Mat. Pura Appl. **200** (2021), 2285–2304; arXiv:2005.13839v2.